

Twistor Encodings as a Conditional High-Coherence Corner of MTT:

Conformal-spin geometry, controlled reduction, and the limits of the bridge

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Abstract

Twistor methods encode important four-dimensional massless, conformal, and self-dual field sectors in holomorphic geometry. This paper asks a narrower MTT question: when may an already selected coherent sector be represented by such twistor data? The answer requires two logically independent bridges. First, controlled projection must reduce upper dynamics to a four-dimensional sector with an explicit error estimate. Second, that sector must carry the conformal spin geometry, integrable twistor distribution, reality structure, and field equations required by the Penrose or Penrose–Ward correspondence. A spectral gap can support the first bridge but cannot create the second. We prove this separation by a finite-dimensional counterexample, derive the exact memory equation for projected dynamics, and show that loss of a gap or error bound does not imply stochastic, set-valued, or kernel-valued evolution. We then state a conditional twistor-descent theorem: if one selected MTT source supplies the projection, four-dimensional conformal-spin carrier, self-dual field sector, twistor double fibration, reconstruction map, and commuting dynamics, the standard twistor correspondence becomes a valid encoding of that MTT sector. Approximate use additionally requires stability estimates for both truncation and holomorphic or self-duality defects. This revision preserves the high-coherence twistor corner as a useful conditional encoding, while withdrawing claims that twistor geometry, optimality, selection events, irreversibility, or arbitrary MTT dynamics follow from spectral suppression alone.

Version 2 Revision Note

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| Supersedes | Version 1.0, released in January 2026. |
| Reason | The previous paper treated spectral-gap reduction as if it generated twistor geometry, called the resulting encoding optimal and faithful without constructing its geometric data, and inferred kernel-valued selection from loss of a truncation bound. |
| Resolution | Version 2 separates dynamical reduction from twistor correspondence, states the full conformal-spin and field-theoretic contract, proves two no-go results for the former implications, and gives an exact conditional descent theorem. |
| Retained content | Schur–Feshbach and memory-kernel control remain useful for estimating leakage from a selected coherent sector. The self-dual massless corner remains a natural place to test a twistor encoding. |
| Open boundary | MTT has not yet selected one upper action and intertwiner that produces the |

physical four-dimensional conformal-spin carrier, its self-dual field equations, twistor reconstruction, coupling normalization, and controlled deformation to the full theory.

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1 The precise question

Twistor theory and MTT address different mathematical operations. Twistor theory replaces suitable four-dimensional fields by holomorphic data on a complex space. MTT proposes that

an effective physical description descends from an upper object through a selected projection. A relation between them therefore has the form

$$\text{upper MTT source} \longrightarrow \text{four-dimensional coherent fields} \longleftrightarrow \text{twistor data}.$$

The left arrow is an MTT reduction problem. The right double arrow is a twistor correspondence problem. The purpose of this paper is to specify when they compose.

The phrase *high coherence* will mean that a declared coherent sector is dynamically stable or approximately autonomous in specified norms. It will not mean holomorphic, conformal, massless, or self-dual unless those properties are separately proved.

1.1 Three status levels

Definition 1.1 (Twistor compatibility). An effective sector is *twistor-compatible* when it has the geometric and field-equation data required by a standard twistor correspondence.

Definition 1.2 (Controlled twistor encoding). A twistor-compatible sector has a *controlled twistor encoding* when the reconstruction map and the projected dynamics commute exactly or up to a stated norm bound on a declared domain.

Definition 1.3 (Selected MTT twistor descent). A controlled twistor encoding is *selected by MTT* only when the projection, conformal-spin geometry, field sector, twistor data, action, normalizations, and error bounds descend from one accepted upper source without being chosen to reproduce the desired correspondence.

The current result reaches the second level conditionally. The third remains an explicit source theorem.

2 Standard twistor data

2.1 Conformal spin geometry

Let $M_{\mathbb{C}}$ be a complex four-manifold, or a suitable complexification of a real four-manifold, with conformal spin structure. Locally its complexified tangent bundle factors as

$$TM_{\mathbb{C}} \cong S \otimes S',$$

where S and S' are rank-two spin bundles. The projectivized primed spin bundle

$$\mathcal{F} = \mathbb{P}(S')$$

is the correspondence space. It carries a rank-two distribution generated locally by

$$V_A = \pi^{A'} \nabla_{AA'},$$

where $[\pi_{A'}]$ is the homogeneous coordinate on the projective spin fiber.

With a fixed chirality convention, integrability of this distribution is tied to vanishing of one half of the conformal Weyl curvature. When the required analytic and regularity hypotheses hold, the local leaf space is a complex three-fold $\mathbb{PT}$, giving the double fibration

$$\begin{array}{ccc} & \mathcal{F} & \\ p \swarrow & & \searrow q \\ M_{\mathbb{C}} & & \mathbb{PT}. \end{array}$$

The sign called self-dual or anti-self-dual varies with orientation and spinor convention. This paper fixes neither name globally; it always means the half selected by the chosen twistor distribution.

In complexified flat spacetime the incidence relation may be written

$$\omega^A = ix^{AA'} \pi_{A'}.$$

This formula depends on an already available four-dimensional spinor factorization. It is not produced by a Hilbert-space projection.

2.2 Field correspondences

The linear Penrose transform identifies suitable sheaf-cohomology classes on twistor space with zero-rest-mass fields on spacetime, subject to domain, support, reality, and regularity hypotheses. In one common helicity convention the relevant group has the form

$$H^1(U, \mathcal{O}(-2h - 2))$$

for helicity h . The precise group changes with helicity convention, region, and support condition.

The nonlinear Penrose–Ward correspondence relates suitable holomorphic vector bundles on twistor space, trivial on the twistor lines $L_x \cong \mathbb{CP}^1$, to local self-dual Yang–Mills connections. The nonlinear graviton construction similarly relates appropriate deformations of twistor complex structure to self-dual conformal geometry. These are substantial geometric theorems, not consequences of a small operator correction.

2.3 Full theories and deformations

It is too strong to say that twistor methods simply fail outside a strictly self-dual sector. Twistor actions and amplitude methods can describe deformations toward full Yang–Mills theory, and several gravitational and supersymmetric sectors have twistor formulations [5, 6, 7]. What changes is the simplicity of the holomorphic correspondence and the data needed to reconstruct a full interacting theory. The MTT claim in this paper is therefore about a particularly transparent corner, not an exclusive domain of all twistor mathematics.

3 What coherent projection does not supply

Proposition 3.1 (Projection and a gap do not create twistor geometry). *A bounded coherent projector and a gapped complementary generator do not imply the existence of a four-dimensional conformal-spin manifold, a twistor space, or a Penrose transform.*

Proof. Take $H = \mathbb{C}^2$,

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad L = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}.$$

The complementary sector is exponentially damped with unit gap and the projected sector is exactly autonomous. These finite-dimensional operator data contain no four-manifold, tangent bundle, conformal class, spin factorization, complex three-fold, incidence relation, or holomorphic bundle. Hence the twistor data do not follow from projection and a gap. $\square$

The counterexample does not deny that a particular MTT source may produce both structures. It shows that the map between them must be an additional theorem.

3.1 Independent control quantities

A meaningful twistor corner needs at least four distinct controls:

$$\varepsilon_{\text{red}} = \text{projected-dynamics or memory error}, \quad (3.1)$$

$$\varepsilon_{\text{geom}} = \text{defect in conformal-spin or twistor-distribution data}, \quad (3.2)$$

$$\varepsilon_{\text{sd}} = \text{self-duality or field-equation residual}, \quad (3.3)$$

$$\varepsilon_{\text{rec}} = \text{reconstruction/intertwining error}. \quad (3.4)$$

Mass and conformal-breaking effects may require additional rows. No theorem presently identifies these quantities with one universal “coherence capacity.” Small ε_{red} does not imply small $\varepsilon_{\text{geom}}$ or ε_{sd} .

4 Controlled reduction of the dynamics

4.1 Exact memory equation

Let $H = H_P \oplus H_Q$, with bounded projection P and $Q = I - P$. For a linear evolution

$$\dot{\Psi} = L\Psi, \quad L = \begin{pmatrix} L_{PP} & L_{PQ} \\ L_{QP} & L_{QQ} \end{pmatrix},$$

write $u = P\Psi$ and $v = Q\Psi$. Whenever variation of constants is valid,

$$\dot{u}(t) = L_{PP}u(t) + L_{PQ}v(t), \quad (4.1)$$

$$v(t) = e^{tL_{QQ}}v_0 + \int_0^t e^{(t-s)L_{QQ}}L_{QP}u(s) \, ds. \quad (4.2)$$

Substitution gives the exact equation

$$\dot{u}(t) = L_{PP}u(t) + L_{PQ}e^{tL_{QQ}}v_0 + \int_0^t K(t-s)u(s) \, ds, \quad K(r) = L_{PQ}e^{rL_{QQ}}L_{QP}. \quad (4.3)$$

This is a reduction identity. It is not yet a twistor statement.

Proposition 4.1 (Gap-controlled memory norm). *Suppose*

$$\|e^{tL_{QQ}}\| \leq M_Q e^{-\lambda_Q t}, \quad t \geq 0,$$

with $\lambda_Q > 0$. Then

$$\|K(t)\| \leq M_Q \|L_{PQ}\| \|L_{QP}\| e^{-\lambda_Q t}$$

and

$$\int_0^\infty \|K(t)\| \, dt \leq \frac{M_Q \|L_{PQ}\| \|L_{QP}\|}{\lambda_Q}.$$

Proof. Apply submultiplicativity to the definition of $K(t)$, then integrate the exponential bound. $\square$

The ratio on the right controls the total memory-kernel norm. Replacing the memory term by a time-local Schur or quasistatic correction requires further resolvent, regularity, and slow-variation hypotheses. The current controlled-truncation paper owns that broader analysis; the present paper uses only the separation in [equation \(4.3\)](#).

4.2 Failure of control is not branching

Proposition 4.2 (Gap loss does not force a kernel transition). *The condition $\lambda_Q = 0$, or failure of a bound based on λ_Q^{-1} , does not imply stochastic, set-valued, or kernel-valued projected evolution.*

Proof. Take any Hilbert space decomposition and let $L = 0$. Then no positive decay gap exists on H_Q . Nevertheless

$$Pe^{tL}\Psi = P\Psi$$

for every t and every Ψ . The projected evolution is the single-valued identity map on H_P . Thus loss of the gap hypothesis does not imply branching. $\square$

More generally, a failed estimate says only that the estimate is unavailable. A non-Dirac reduced probability kernel requires a source ensemble or state, disintegration along fibers, and actual variation of projected outcomes within a fiber. An MTT selection event additionally requires a selected branch or instrument law. None follows from $\lambda_Q \rightarrow 0$.

5 The conditional MTT-to-twistor bridge

5.1 The source contract

Definition 5.1 (Selected twistor source contract). A selected MTT twistor descent on a slab consists of:

1. an upper state complex or field space $\mathcal{H}_{\text{up}}$ and selected action or generator L_{up} ;
2. a coherent projector P , domain, norms, and reduction estimate;
3. a four-dimensional oriented conformal-spin carrier $(M, [g], S, S')$, including signature, complexification, and reality data;
4. a field map $\mathcal{J} : \text{Ran}P \rightarrow \mathcal{E}(M)$ into a declared gauge, gravity, or zero-rest-mass sector;
5. the self-dual or conformal field equation obeyed by $\mathcal{J}(P\Psi)$, exactly or with a residual bound;
6. the correspondence space $\mathcal{F} = \mathbb{P}(S')$, twistor distribution, leaf space $\mathbb{PT}$, and double fibration;
7. a Penrose, Penrose–Ward, or nonlinear-graviton transform $\mathcal{W}$ with its bundle, line-triviality, cohomology, and reality conditions;
8. an intertwining relation between upper, spacetime, and twistor dynamics;
9. an action and norm bridge fixing couplings and positive physical inner products;
10. separate error bounds for reduction, geometry, self-duality, and reconstruction.

The contract deliberately does not include a universal threshold at which twistor data “become a kernel.” If the contract fails, the conclusion is that this encoding is no longer certified.

Theorem 5.2 (Conditional twistor descent). *Assume all exact rows of Definition 5.1 on a common domain, and assume the chosen standard twistor correspondence is bijective on the declared quotient or gauge-equivalence classes. Then*

$$\mathcal{T} = \mathcal{W} \circ \mathcal{J} \circ P$$

is a well-defined twistor encoding of the selected coherent MTT sector. If the dynamics intertwine,

$$\mathcal{T}e^{tL_{\text{up}}} = U_t^{\text{tw}}\mathcal{T}$$

on that sector, the spacetime and twistor descriptions represent the same reduced evolution there.

Proof. The projector sends admissible upper states to the declared coherent domain. The field map sends that domain to solutions of the field equation for which the chosen twistor correspondence is defined. The correspondence therefore assigns a unique twistor equivalence class. Composition gives $\mathcal{T}$. The displayed intertwining relation identifies the time evolution of the two encodings. Gauge or cohomological equivalence is handled by the quotient included in the hypotheses. $\square$

This theorem is a composition theorem, not a derivation of the standard Penrose–Ward correspondence. Its MTT content is the demand that all rows come from one source and commute.

5.2 Controlled approximate use

Suppose instead that the four maps commute only approximately. Let $\|\cdot\|_{\text{tw}}$ be a norm on a fixed gauge slice or quotient chart where reconstruction is locally Lipschitz with constant $C_{\mathcal{W}}$. If

$$\varepsilon_{\text{tot}} = C_{\mathcal{W}}(\varepsilon_{\text{red}} + \varepsilon_{\text{geom}} + \varepsilon_{\text{sd}} + \varepsilon_{\text{rec}})$$

is finite on a controlled domain, then the twistor representation has at most that declared error. This estimate is conditional on the local stability of the transform. Small curvature residual alone does not prove such stability; gauge fixing, elliptic estimates, and possible cohomological obstructions must be controlled.

6 Gauge and gravity examples

6.1 Self-dual Yang–Mills

For a gauge bundle $E \rightarrow M$, the transparent classical target is

$$F_A^- = 0$$

in a fixed chirality convention. The Ward transform assigns a holomorphic bundle $\mathcal{E} \rightarrow \mathbb{PT}$, trivial on each twistor line, to a gauge equivalence class of local self-dual connections. In an MTT realization the following are still separate obligations:

1. show that $\mathcal{J}(P\Psi)$ is a connection on E ;
2. derive $F_A^- = 0$ or bound its residual from the upper action;
3. prove line triviality and the required reality condition;
4. match the twistor and four-dimensional action normalizations;
5. control the deformation toward full Yang–Mills.

The existing Theta III paper studies the fourth row conditionally. It explicitly assumes the twistor corner and reconstruction-and-norm bridge. That result is compatible with this paper but does not select the bridge.

6.2 Self-dual conformal gravity

For gravity, the nonlinear-graviton route additionally requires a four-dimensional conformal structure whose selected Weyl half vanishes and a family of suitable rational curves in the twistor space. An upper metric Hessian or a stable coherent subspace does not by itself provide those curves. To connect this route to the MTT gravity program one must show that the selected effective metric, its spin connection, and its self-dual curvature arise from the same upper action that controls the coherent projection.

6.3 Mass, non-self-duality, and generic backgrounds

Massive fields, both helicity sectors, generic curvature, and full interactions do not fit the simplest Penrose or Ward correspondence. Possible extensions include ambitwistor constructions, twistor actions with additional nonlocal terms, deformations of holomorphic structures, and spinor-helicity or amplitude formulations. Choosing among them is a model decision. MTT has not proved that one particular extension is selected by coherence, nor that increasing mass or anti-self-dual curvature produces a universal transition at one scalar threshold.

7 Determinism, probability, and selection

Let $\Pi : X \rightarrow Y$ be any measurable projection and $T_t : X \rightarrow X$ an upper evolution. A deterministic reduced map $F_t : Y \rightarrow Y$ exists exactly when

$$\Pi T_t(x_1) = \Pi T_t(x_2) \quad \text{whenever} \quad \Pi x_1 = \Pi x_2$$

on the declared domain. This is fiber invariance. It is an algebraic condition, not a spectral-gap condition.

If fiber invariance fails and a probability measure on X is supplied, disintegration can produce a conditional probability kernel on Y . The kernel depends on that measure. It is not generated by noninjectivity alone. Selecting one physical outcome requires still more: an instrument, stopping law, recorder, or other accepted branch rule.

Consequently, a breakdown of the twistor encoding has only the following immediate meaning:

the hypotheses or error tolerance of this representation have failed.

It does not prove irreversibility, randomness, collapse, a new basin, or an arrow of time.

8 Relation to the MTT corpus

8.1 Foundation and fixed points

The Foundation and Fixed Points papers provide the upper vocabulary of selection, projection, stability, and conditional continuation. They do not by themselves select a four-dimensional twistor double fibration. This paper uses their projection language only on a declared slab.

8.2 Controlled truncation

The controlled-truncation paper supplies the proper operator-theoretic setting for spectral projectors, resolvents, Schur complements, and finite-time error. The memory equation in

this paper is a short interface to that result. Twistor geometry enters only after the separate geometric rows of Definition 5.1.

8.3 Theta closure

Theta I and Theta III use a twistor corner as a computational and normalization device. Their revised form treats self-dual dynamics, reconstruction, fiber measure, and action normalization as assumptions or conditional bridges. The present paper agrees with that status. It neither duplicates their coupling calculations nor promotes them to a source theorem.

8.4 Protospinor and shared-circle programs

Protospinor and shared-circle constructions may help build the conformal spin and phase data needed by rows 3 and 6. Equality of ranks or reuse of a common $U(1)$ line is not enough. The decisive map must preserve the connection, chirality, incidence relation, and field operator through projection.

9 What is achieved and what remains

Layer	Current result	Remaining source obligation
Coherent reduction	Exact memory identity and conditional gap bound	Selected physical generator, domain, and error certificate
Twistor geometry	Standard conformal-spin contract stated	Derive the carrier, distribution, reality structure, and leaf space
Gauge correspondence	Standard Ward route identified	Prove the selected coherent field is in the required self-dual sector
Gravity correspondence	Nonlinear-graviton route scoped	Derive the effective conformal geometry and rational-curve family
Action and norms	Kept separate from holomorphic field equations	Same-source positive norm and coupling normalization
Breakdown	Gap-loss implication disproved	Model-specific continuation or selection law, if one exists

The most useful next theorem is not that “high coherence becomes twistor space.” It is a commuting construction of rows 2–8 of Definition 5.1 for one selected gauge or gravity sector. A successful first example could be local and self-dual. It should publish the upper source, the four-dimensional spin bundle, the twistor distribution, the field map, the Ward bundle, and one nontrivial intertwining check.

10 Falsifiability

For any proposed MTT twistor realization, one should be able to test:

1. whether the effective tangent bundle actually factors as required;
2. whether the twistor distribution is integrable on the stated domain;

3. whether the projected field satisfies the target equation;
4. whether the holomorphic bundle is trivial on twistor lines;
5. whether reconstruction returns the original gauge class;
6. whether the action and norm coefficients agree without target fitting;
7. whether the four error rows remain below their declared tolerances.

Failure of any row falsifies that encoding on the tested domain. It does not falsify twistor theory or MTT globally.

11 Conclusion

Twistor theory can be a powerful language for a selected MTT coherent sector, but only after the relevant four-dimensional conformal-spin and field data exist. Spectral projection and Schur–Feshbach control address the stability of a reduction. Penrose, Ward, and nonlinear-graviton correspondences address the holomorphic representation of special four-dimensional equations. These are complementary bridges, not interchangeable consequences.

Version 2 therefore retains a conditional high-coherence twistor corner and makes its content exact. It proves that projection and a gap do not generate twistor geometry, and that loss of a gap bound does not force branching. It then identifies the constructive route forward: one selected upper action must emit the coherent projector, conformal-spin carrier, self-dual field sector, twistor correspondence, reconstruction, normalization, and error control in a commuting diagram. Until that theorem is supplied, the twistor corner is a disciplined encoding proposal rather than a derived universal limit of MTT.

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